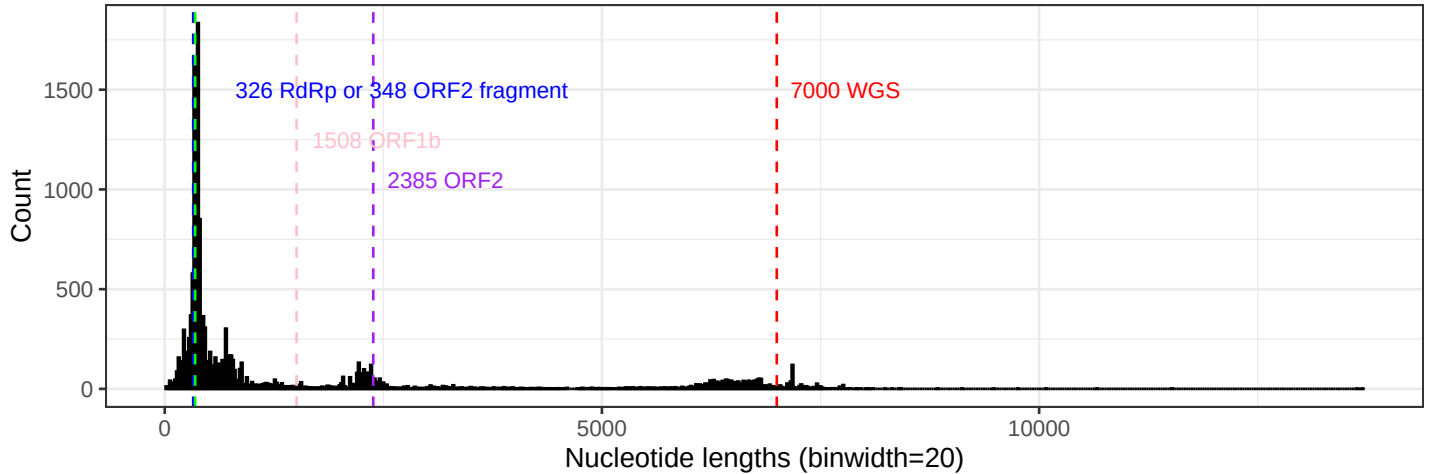


Exploratory Graphics

What is the genome length of astrovirus?

The astrovirus genome is approximately 6 to 8kb in length.

Diagnostic plot to estimate min-length filter for phylogenetic analysis (all data = 16381)

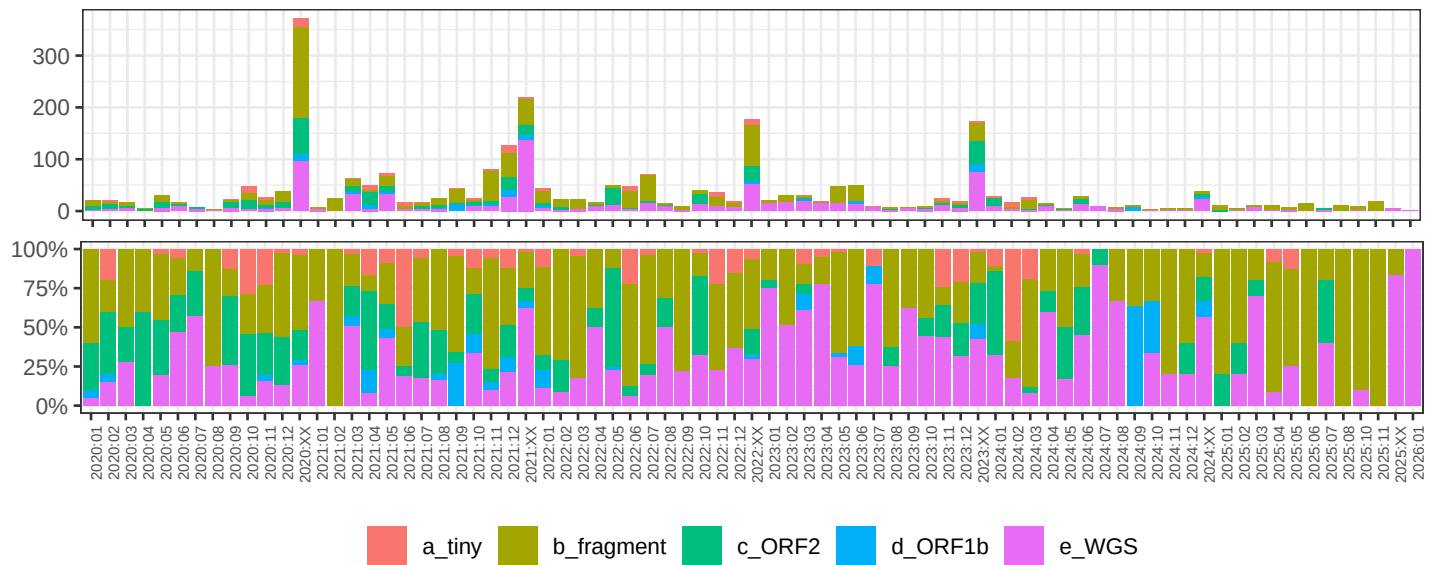


What is the distribution by length and host type?

```
cdata <- data %>%
  subset(., !is.na(date)) %>%
  subset(., date != "XXXX-XX-XX") %>%
  mutate(
    date_adjusted = lubridate::date(gsub("-XX", "-01", date)),
    col_year = substr(date, 1, 4),
    col_mon = substr(date, 6, 7),
    col_day = substr(date, 9, 10),
    len_category = case_when(length > 4000 ~ "e_WGS",
                             length > 1700 ~ "c_ORF2",
                             length > 1000 ~ "d_ORF1b",
                             length > 300 ~ "b_fragment",
                             TRUE ~ "a_tiny"),
    host_category = case_when(host %in% c("Homo sapiens") ~ "Z_Human",
                              is.na(host) ~ "NA",
                              TRUE ~ "A_Other"),
    host_length = paste0(host_category, "_", len_category)
  )

diagnostic_time_fill_plot_month(
  subset(cdata, col_year > 2019),
  margin_l=0.5, fill="len_category"
)
```

Frequency and proportion (n=2731)

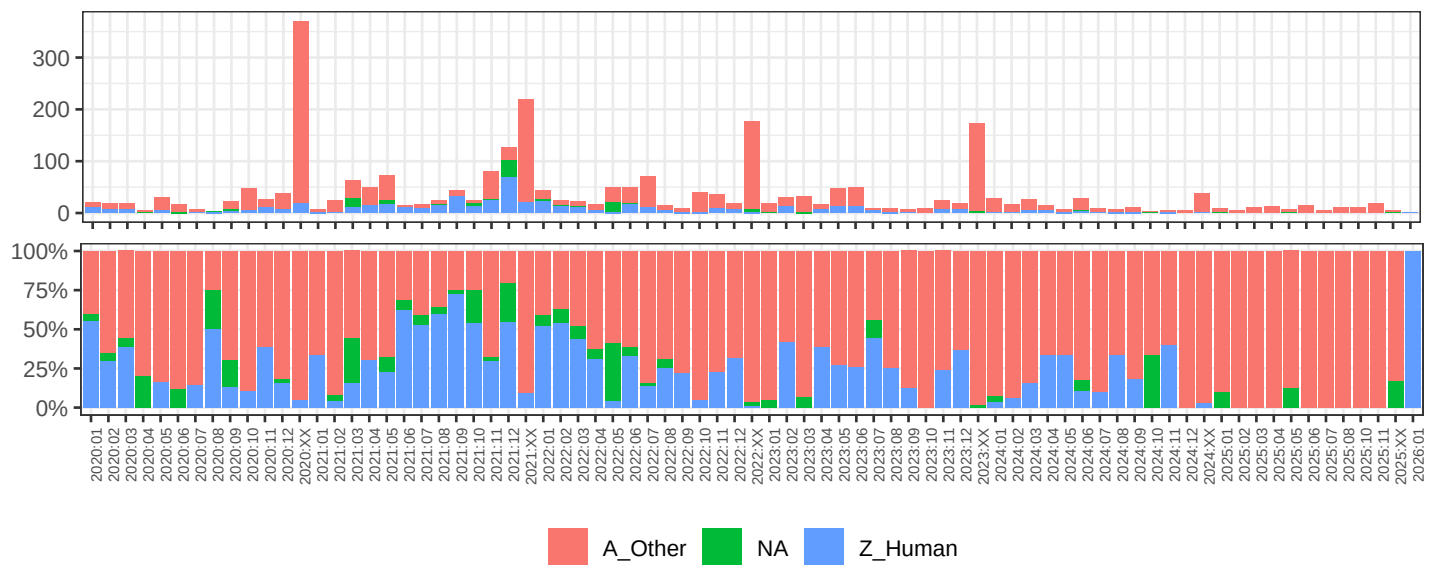


```
cdata %>% filter(col_year > 2019) %>% group_by(len_category) %>% summarise(n = n(), .groups = "drop")
```

```
## # A tibble: 5 x 2
##   len_category    n
##   <chr>        <int>
## 1 a_tiny       185
## 2 b_fragment   1103
## 3 c_ORF2       484
## 4 d_ORF1b     123
## 5 e_WGS       836
```

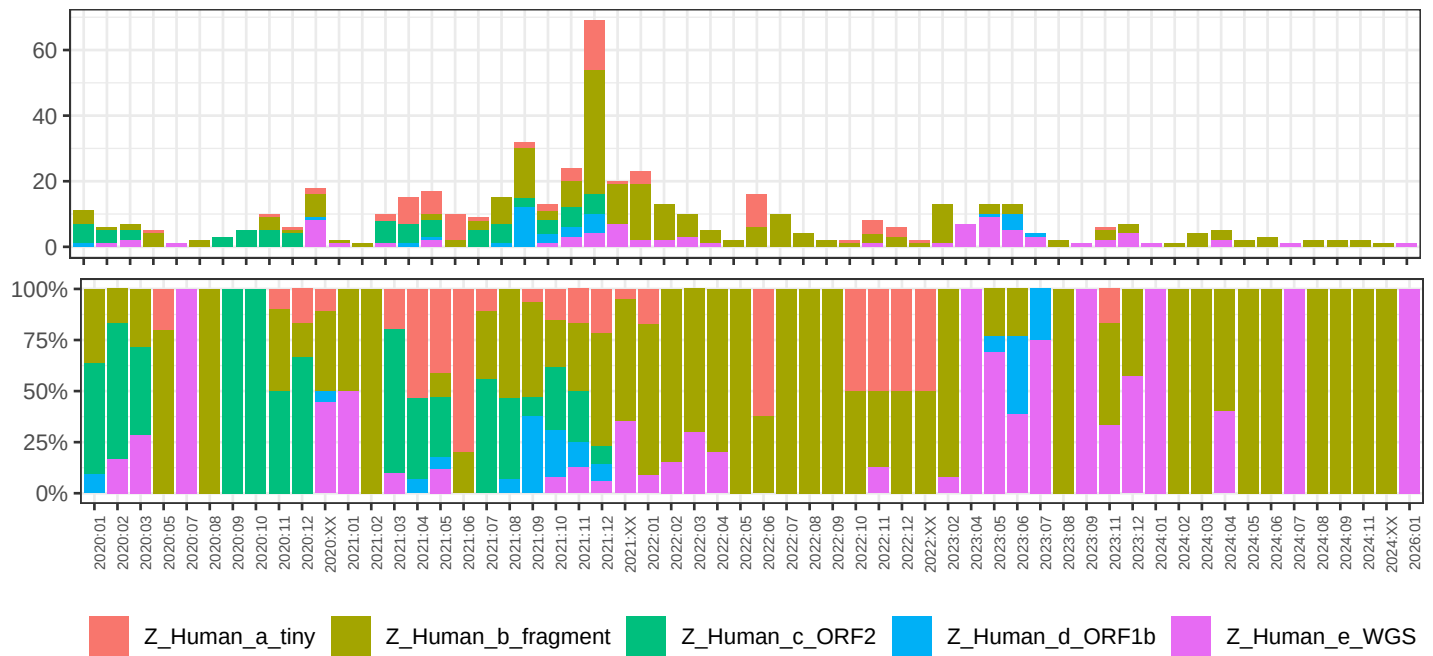
```
diagnostic_time_fill_plot_month(
  subset(cdata, col_year > 2019),
  margin_l=0.5, fill="host_category"
)
```

Frequency and proportion (n=2731)



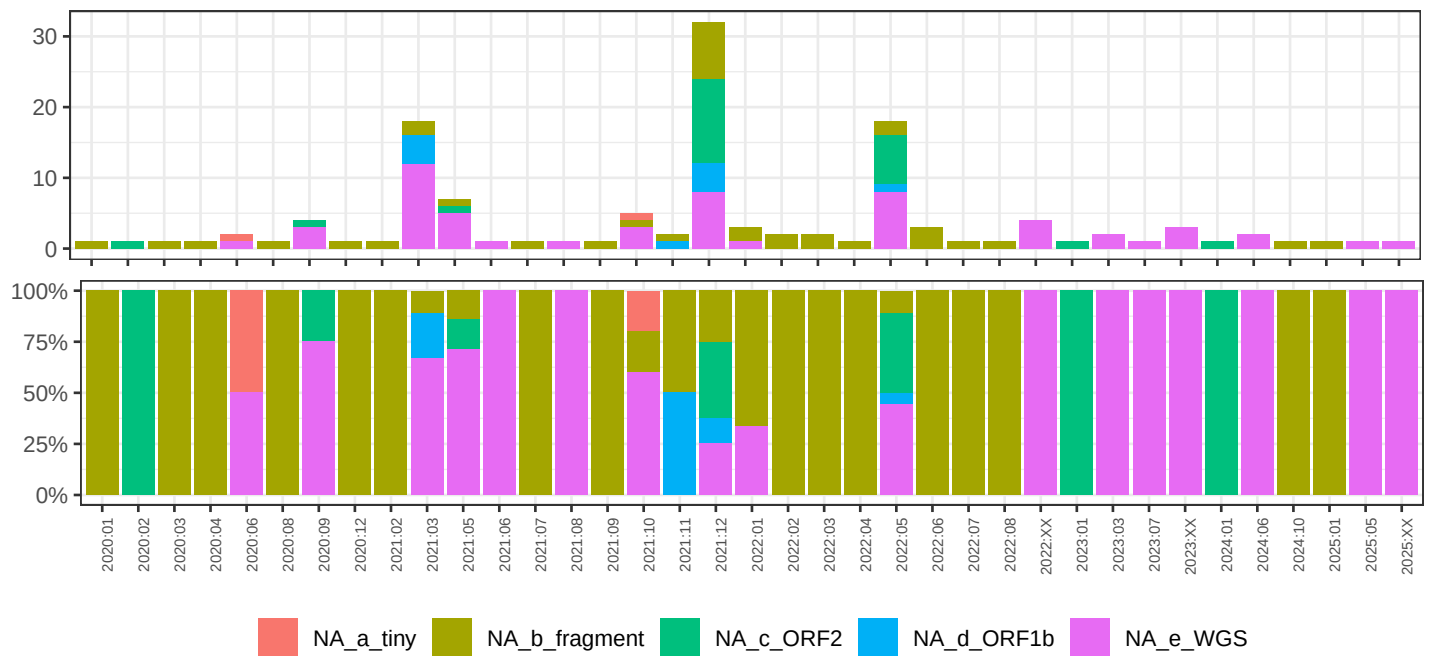
Focused on human samples:

Frequency and proportion (n=505)

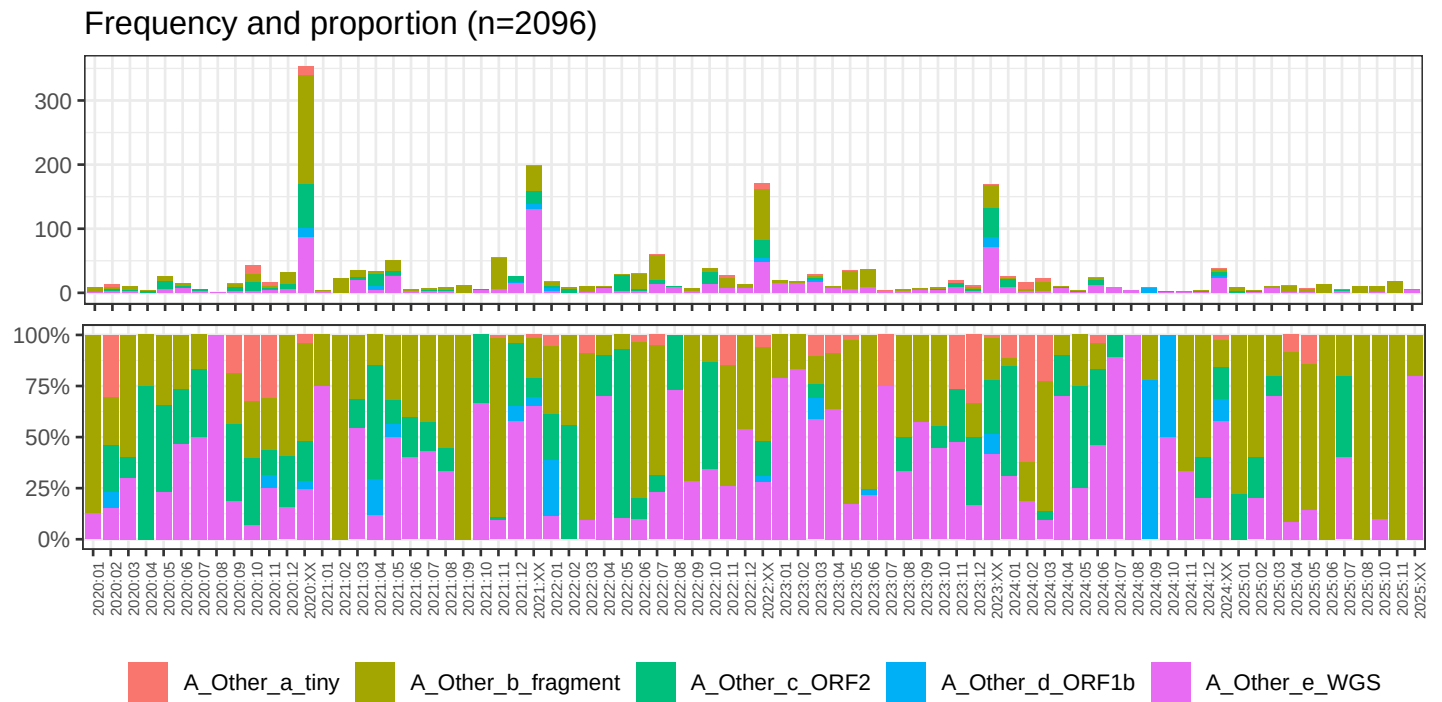


Focused on "NA" samples:

Frequency and proportion (n=130)



Focused on "Other" samples:



Who are the main submitters of sequence data?

Table 1: Top 25 most frequent sequence submitters with their region and countries

authors	n	regions	countries
?	750	Asia, North America, NA, Africa, South America	China, USA, NA, Malaysia, Bangladesh, Republic
tao et al.	709	Asia	China
armero et al.	425	Asia	China, Vietnam
li et al.	340	Asia, North America	China, USA, Canada
chen et al.	323	Asia	China
wang et al.	311	Asia, Africa, North America, NA	China, Kenya, Uganda, USA, NA
qiu et al.	302	Asia	China
wu et al.	259	Asia	China, Thailand, Taiwan
zhou et al.	257	Asia, Europe	China, Spain
he et al.	254	NA, Asia	NA, China, Myanmar
niendorf et al.	205	Europe	Germany, Hungary, Czech Republic
chu et al.	199	Asia	Hong Kong, Vietnam, Sri Lanka, Cambodia
yoshitomi et al.	187	Asia	Japan
pativada et al.	181	Asia	India
lee et al.	174	Asia, NA	South Korea, NA
zhirakovskaia et al.	166	Europe	Russia
yan et al.	165	Asia	China
zhang et al.	162	Asia, North America, Africa	China, USA, Sierra Leone
ushijima et al.	160	Asia	Japan
jindal et al.	158	North America	USA
zhu et al.	157	Asia, NA	China, NA
chidoti et al.	155	Africa	Zimbabwe
sharp et al.	147	Europe	United Kingdom
guix et al.	145	Europe, NA, Asia	United Kingdom, NA, Spain, Albania, France, Net
wei et al.	144	Asia	Thailand, China

Table 2: Top 25 most frequent sequence submitters with their region and countries

authors	n	regions	countries
homo sapiens	4741	Europe, Asia, North America, Africa, South America, NA, Oceania	Italy, China, Thailand
NA	2222	Asia, Africa, Oceania, NA, North America, Europe, South America	China, Cameroon, N
sus scrofa	1462	Asia, South America, Europe, North America, Africa, Oceania, NA	China, Chile, Hunga
gallus gallus	765	Asia, Europe, Africa, South America, North America, NA, Oceania	China, Belgium, Jap
sus scrofa domesticus	560	Oceania, Africa, Europe, Asia, North America	Australia, South Afr
rattus norvegicus	537	Europe, Asia, Africa	Germany, Hungary,
bos taurus	500	Asia, Europe, North America, Oceania, Africa, South America	South Korea, Iran, C
meleagris gallopavo	400	NA, South America, Africa, North America, Europe	NA, Ecuador, Nigeri
anatidae	366	Asia, NA, Europe, North America	China, NA, Croatia,
canis lupus familiaris	347	Europe, South America, Asia, Oceania, North America	Greece, Hungary, Br
miniopterus schreibersii	207	Asia, Europe	China, Italy, South I
felis catus	182	Asia, North America, Oceania, Europe	Hong Kong, USA, A
miniopterus magnater	150	Asia	Hong Kong, Thailand
taphozous melanopogon	134	Asia	China, Thailand, Ca
mus musculus	130	North America, Asia, Africa, South America, Europe	USA, China, Madag
capra hircus	103	Asia, Europe, Africa, North America	China, Hungary, Ga
macaca mulatta	96	Asia, NA	Bangladesh, NA
rattus rattus	93	Africa, Asia	Madagascar, Réunio
aves	92	Asia, North America	China, India, USA
hipposideros	90	Africa, Asia	Zimbabwe, China, R
mormopterus francoismoutoui	84	Africa	Réunion
mus musculoides	84	Africa	Gabon
rattus tanezumi	84	Asia, Africa	China, Republic of t
columba livia	82	Asia, Africa, Europe	Kazakhstan, Réunio
eothenomys cachinus	75	Asia	China

What is the host/composition in each cluster?

major_cluster2	cluster_size	hosts
AStV-2	214	Anatidae(195), NA(7), Anser anser(5), Anser cygnoides(3), Anser sp.(2), Gallus gallus(2)
AStV-1	209	Homo sapiens(170), NA(39)
AStV-3	92	Sus scrofa(74), NA(9), Sus scrofa domesticus(6), Bos taurus(1), Canis lupus familiaris(1), Rattus norvegicus(1)
AStV-4	81	Sus scrofa(53), NA(10), Camelus dromedarius(8), Sus scrofa domesticus(7), Camelus bactrianus(2), Ailurus fulgens(1)
AStV-5	73	Canis lupus familiaris(65), NA(4), Nyctereutes procyonoides(2), Canis lupus(1), Cerdocyon thous(1)
AStV-6	67	Bos taurus(59), Bos grunniens(2), Budorcas tibetana(2), Capreolus capreolus(2), Bos mutus(1), Canis lupus familiaris(1)
AStV-7	61	Gallus gallus(53), Neogale vison(5), Canis lupus familiaris(2), Ptyas mucosa(1)
AStV-8	58	Sus scrofa(40), Sus scrofa domesticus(18)
AStV-9	44	Felis catus(38), Feliformia(2), Neogale vison(1), Nyctereutes procyonoides(1), Panthera tigris altaica(1), NA(1)
AStV-10	31	Anatidae(20), Anser sp.(7), Anser(2), NA(2)
AStV-11	30	Gallus gallus(21), NA(6), Neogale vison(2), Sus scrofa(1)
AStV-13	27	Anatidae(11), Cairina moschata(11), NA(4), Anas platyrhynchos(1)
AStV-12	22	Sus scrofa(21), Camelus bactrianus(1)
AStV-15	20	Ovis aries(11), Capra hircus(9)
AStV-16	19	Anatidae(17), Anas platyrhynchos(1), NA(1)
AStV-17	18	Bos taurus(17), Bos mutus(1)
AStV-14	17	Sus scrofa(16), Sus scrofa domesticus(1)
AStV-18	13	Mus musculus(8), NA(5)
AStV-19	12	Sus scrofa(6), Sus scrofa domesticus(4), NA(2)
AStV-20	12	Homo sapiens(7), NA(5)
AStV-21	11	Rattus norvegicus(4), Rattus andamanensis(2), Suncus murinus(2), Mus musculus(1), Rattus tanezumi(1), NA(1)
AStV-23	11	NA(11)
AStV-22	10	Bos taurus(6), Canis lupus familiaris(2), Bos mutus(1), Bubalus bubalis(1)
AStV-25	10	Bos taurus(6), Capra hircus(3), Bos grunniens(1)
AStV-26	9	Sus scrofa(4), Sus scrofa domesticus(3), NA(2)
AStV-27	9	Rattus norvegicus(6), Rattus nitidus(1), Suncus murinus(1), Sus scrofa(1)
AStV-24	8	Sus scrofa(7), Sus scrofa domesticus(1)
AStV-28	8	Columbidae(3), Spilopelia chinensis(3), Columba livia(1), NA(1)
AStV-29	8	Capra hircus(6), Ovis aries(1), NA(1)
AStV-30	7	Anatidae(5), Anas platyrhynchos(1), Anser anser(1)
AStV-31	7	Felis catus(3), Ailurus fulgens(1), Canis lupus familiaris(1), Nyctereutes procyonoides(1), NA(1)
AStV-32	7	Ovis aries(3), Bos taurus(2), NA(2)
AStV-33	6	Homo sapiens(5), NA(1)
AStV-35	6	Neogale vison(4), Meleagris gallopavo(2)
AStV-36	6	Marmota caudata(6)
AStV-34	5	Rattus andamanensis(4), Suncus murinus(1)
AStV-37	5	Homo sapiens(3), NA(2)
AStV-38	5	Ovis aries(4), Ovibos moschatus(1)
AStV-40	5	Pipistrellus abramus(2), Myotis davidii(1), Myotis laniger(1), Myotis sp.(1)
AStV-39	4	Bubalus bubalis(3), Bos mutus(1)
AStV-41	4	Bos taurus(4)
AStV-42	4	Oryctolagus cuniculus(4)
AStV-43	4	Zalophus californianus(4)
AStV-44	4	NA(4)
AStV-45	4	Marmota himalayana(2), Marmota caudata(1), Marmota sibirica(1)
AStV-46	4	Marmota himalayana(4)
AStV-47	4	Marmota himalayana(4)
AStV-48	4	Rattus tanezumi(3), Paguma larvata(1)
AStV-49	4	Canis lupus familiaris(4)
AStV-50	4	Hipposideros armiger(2), Rhinolophus sp.(2)
AStV-51	4	Miniopterus schreibersii(2), Chiroptera sp.(1), Miniopterus fuliginosus(1)
AStV-52	4	Hipposideros larvatus(3), Hipposideros armiger(1)
AStV-53	3	Homo sapiens(3)
AStV-54	3	NA(2), Mustela putorius furo(1)
AStV-56	3	Arnoglossus sp.(1), Echelus uropterus(1), NA(1)
AStV-57	3	Aves(3)
AStV-58	3	Phrynocephalus erythrurus(3)
AStV-59	3	Canis lupus familiaris(3)
AStV-60	3	Berylmys bowersi(1), Rattus losea(1), Rattus tanezumi(1)
AStV-61	3	Miniopterus sp. B76(2), Miniopterus minor(1)
AStV-62	3	Neodon fuscus(3)

